

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year: 2019- 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. Mechanical Engineering

Course Code & Title: ME8694-Hydraulics and Pneumatics

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Innovative practice: Model Showcase

Topic: Actuators and Control Valves

Actuators and its classification, various valves like directional control valve, pressure valve, flow control valves are explained inside the classroom by showcasing the original component. The component are looks similar by seeing the ANSI symbol engraved in the item we can able to identify the type of valves. The components which are showcased inside the classroom are listed below

- Single Acting Cylinder with spring return (Actuator)
- Double Acting Cylinder (Differential type) (Actuator)
- 3/2 Push button operated Directional Control Valve (Valve)
- 5/2 Double pilot operated Directional Control Valve (Valve)
- 5/2 Double Solenoid Directional Control Valve (Valve)
- One-way flow control valve (Valve)
- Twin Pressure valve or 'AND' function (Valve)
- Shuttle valve or 'OR' function (Valve)
- Pressure Sequence valve (Valve)

The above components are handover to the students at the end of the session. Finally, questions are asked to the students to identify the type of component. All students are answered correctly. Teaching by showing the real component is very useful and interactive learning for this subject. The components are taken from Mechatronics Laboratory. Sample photographs are taken during the session.

DURING MODEL SHOWCASE SESSION



Student clarified the doubt about the component



Faculty member raised a question to the student to identify the component



Faculty member raised a question to the student to identify the component



Faculty member explained about 5/2 double pilot operated DCV to the students
SESSION OUTCOMES:

- This makes the students to learn the concept more effectively.
- It gives confidence to select and place the component in an automation industry.
- This showcase activity provides the student to remember the type of components.

CO-PO Mapping

Course Outcome/ Program Outcome	PO2	PO3	PO4	PO5	PO9	PO10
CO2- Summarize the features and functions of Hydraulic motors, actuators and Flow control valves	2	2	2	2	2	2